CSC 212—Algorithms and Complexity
Create a Java program to do x^e for very large numbers.
Practical 4 Term 4 26 September 2016

Due by 23h59 on 2 October 2016

Implement code to: (1) Read in very large numbers of up to 30 digits. (2) Implement the arithmetic operations: addition, subtraction, and multiplication, modulus, division and exponentiation. (3) Implement the exponential operation x^e between any two of these 30 digit numbers and give the result accurate to the necessary number of digits. Limit e to 4 digits. This means that the number $(10^{30} - 1)^{9999}$ requires about three hundred thousand digits must be calculated accurately.

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1. In your home directory, inside your `surname,firstname` directory ensure that you have a directory called `44practical`. Submit this practical as usual by entering `cd; make submit` from the CLI.

2. Run your code using a `Makefile` that passes pairs of numbers separated by a symbol for the arithmetic operation that needs to be done. Note this is a very primitive interface to make it possible to mark the code easily by using a batched script.

3. Call your executable java `calculate` and run it as follows:
   ```
   java calculate 123456 + 1234
   java calculate 123456 - 1234
   java calculate 123456 * 1234
   java calculate 123456 / 1234
   java calculate 123456 % 1234
   java calculate 123456 ^ 1234
   ```
   Note that the program takes two numeric strings as operands separated by an operation code, i.e., three arguments altogether. You might encounter problems reading the data from the command line exactly as given above. Print only the last 10 digits of your results.

4. Note that this practical is not as trivial as it seems.

5. Handing in code working code for this practical is critical for this practical you will not be allowed to write the examination for CSC212. Handing in is done with the usual `cd; make submit` process. The `Makefile`s are in `/export/home/notes/ds/makefiles/`. The one goes into your home directory and the other must be in your current practical's directory, i.e., `44practical`.
